Revision History

Version	Date	Description	Author(s)	Reviewer(s)
v1.1	2025-11-18	Initial Phase 1 draft: purpose, personas, scenarios, CQs	Sainbuyan L., Munkhnaran B.	Altangerel
v1.2	2025-11-25	Finalized Phase 1 for submission	Sainbuyan L., Munkhnaran B.	Altangerel
v2.1	2025-12-03	Phase 2: Information gathering, methodology	Munkhnaran B., Altangerel	Sainbuyan
v2.2	2025-12-03	Finalized Phase 2 for submission	Munkhnaran B., Altangerel	Sainbuyan

Contributions

- Sainbuyan L.: Informal purpose, personas, scenarios drafting.
- Munknaran B.: Concept identification and competency questions.
- Altangerel N.: Relationships, ER conceptual model and focus classification.
- Sainbuyan L. (Project Manager): Consolidation, editing and submission on behalf of the team.

Knowledge Graph of Mongolian Cinema

Team-1

December 2025

PHASE 1 — Purpose Definition

1. Informal Purpose

When searching for information about Hollywood or Korean films, K-drama online, users can easily access comprehensive information in one place—release year, director, production company, genre, rating, viewer reviews, cast, and their related works. This is possible because these industries have established interconnected structured, knowledge ecosystems supported by mature knowledge graphs.

In contrast, the Mongolian film domain currently lacks such a unified, linked data ecosystem. Information about Mongolian cinema is scattered across many sources, unstructured, inconsistent and disconnected, difficult to integrate or analyze in a single search. Therefore, this project aims to:

- consolidate dispersed and unstructured Mongolian cinema data,
- build a connected data model using ontology and knowledge-graph methodology,
- provide a comprehensive, searchable, analyzable Knowledge Graph of Mongolian Cinema.

This knowledge graph will enable researchers, film enthusiasts, and media organizations to:

- explore connections among Mongolian films,
- analyze relationships between filmmakers, actors, and studios,
- study genres, award trends, and historical developments,
- access all essential information about a film from a single query,
- perform recommendations and graph-based analytics.

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2. Personas

Persona 1: Non-Fan Viewer
Goal: Has no film knowledge. Wants to pick a high-rated film by a director/actor his girlfriend likes for a movie date.
Needs:

- View linked information among director, film, actors
- Runtime and short summary
- High-rated films featuring a preferred artist

Persona 2: Film Enthusiast

Goal: Find other works by a favorite actor or director.
Needs: Similar-genre films, related works by the same creators, ability to discover underrated, lesser-known but high-rated films. Importance placed on rating and release year.

Persona 3: Business Owner

Goal: Collaborate with filmmakers popular among target customers (e.g., children, women).
Needs: Networks of artist collaborations, film ratings and recent releases, identify current trending filmmakers with strong portfolios.

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3. Scenarios

Scenario A: Non-Fan Viewer

Selecting a good Mongolian film for a date.

- **KG Needs:** Actor - list of other high-rated films; Movie - synopsis, runtime; Director – genre connections to understand film type.

Scenario B: Film Enthusiast

Finding underrated films by a favorite director.

- **KG Needs:** Director B – list of all directed films; Filter by high rating, low review count; Release year is before 2000.

Scenario C: Business owner

Collaborate with currently active, trending filmmakers.

- **KG Needs:** Production company - list of produced films; Filter by 2020–2024 high-rated releases; Identify frequently collaborating actors and directors.

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4. Competency Questions (CQs)

The knowledge graph must be able to answer:

1. Can Mongolian films be listed by genre?
2. What films has Director X made, and in which years?
3. Which actors starred in Movie Y?
4. Which Mongolian actors have the highest number of film appearances?
5. Which Mongolian films have won domestic or international awards?
6. What works has Production Company Z released?
7. What common features link films of a specific genre?
8. Can we generate an actor-actor or actor–director collaboration network?
9. Who are the trending filmmakers of the last five years?
10. Can we visualize the career timeline of an actor or director?

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5. Concept Identification

5.1. Core Concepts

Table 1: Concept Definition

Concept	Description
Movie	Metadata including release year, runtime, genre, rating, summary, awards, review co
Person	General category for creative individuals (actors, directors, producers, cinematograp
Actor	Subtype of Person. Performs roles in films.
Director	Subtype of Person. Oversees artistic and narrative aspects of a film.
Production Company	Entity responsible for financing, production, or distribution.
Genre	Category such as drama, comedy, action, arthouse, etc.
Rating	Numerical quality score (e.g., IMDB score, viewer rating).
Review Count	Number of user reviews or views (used for detecting underrated films).
Release Year	Year the film premiered.
Collaboration	Repeated or significant cooperation between artists (useful for Persona 3).

Table 2: Relationships

Relationship	From	To
Acted in	Actor	Movie
Directed by	Movie	Director
Produced by	Movie	Production Company
Has genre	Movie	Genre
Released in	Movie	Release Year
Has rating	Movie	Rating
Has reviews	Movie	Review Count

6. Relationships

7. ER Diagram (Conceptual Model)

Entities:

- Movie: ID, Name, Release Year, Type, Genre, Rating, Runtime
- Actor: ID, Name, Birth date, Profession
- Director: ID, Name
- Production: ID, Name

Relationships:

- Movie → Actor (`acted_in`)
- Movie → Director (`directed_by`)
- Movie → Production (`produced_by`)

8. Focus Classification

9. PF-Sheet

PHASE 2 — Information Gathering

In this section, the second main input of the project is described, namely the list of data sources and datasets available for building the Knowledge Graph of Mongolian Cinema. The objective of this phase is to collect and organize the raw knowledge needed to build a unified, structured representation of Mongolian cinema data. Unlike Hollywood or Korean cinema ecosystems,

Table 3: Focus Classification

Concept	Focus Level
Movie	High
Actor	High
Director	High
Genre	Medium
Rating / Review Count	High
Production Company	Medium

Aspect	Definition
Domain	Mongolian cinema (films, cast, crew, studios, genres, ratings)
Problem	Information is scattered, unstructured, and lacks interconnection
Objective	Create a connected, structured knowledge graph integrating all relevant film data
Scope	Mongolian films with metadata: title, release year, director, cast, production company, ger
Expected Users	Researchers, film enthusiasts, media organizations
Output	Knowledge graph (movies, actors, directors, production; edges: acted_in, directed_by, pro

which already benefit from mature, interconnected knowledge graphs that allow users to easily find release year, director, production company, genre, rating, reviews, and filmographies in one place, the Mongolian film domain lacks such a centralized knowledge infrastructure. Our goal is to gather, clean, standardize, and prepare these resources to enable the creation of such a connected ecosystem.

1. Sources Identification

- **Wikipedia Mongolia (mn.wikipedia.org)** Provides semi-structured pages for individual Mongolian films, including titles,directors, actors, production companies, runtimes, summaries, and award information. Type: Language + Data-value resource. Diversity Awareness: Not diversity-aware; structure varies between pages.

- **Kinosan.mn** A local Mongolian online movie portal listing films, posters, cast, genre tags, summaries, and sometimes ratings and release years. Type: Data-value resource. Diversity Awareness: Not diversity-aware; heterogeneous formats.

- **Domain Vocabulary Notes** Informal text resources describing film concepts (Movie, Person, Actor, Director, Genre, Rating). Type: Language resource. Diversity Awareness: Yes; domain vocabulary structures are fully under project control.

2. Datasets Collected

The following datasets were gathered and consolidated during the information gathering phase:

- **Movie Dataset** Contains metadata for Mongolian films:
  - Title, Release Year, Runtime
  - Genre
  - Rating
  - Actors
  - Director
  - Production Company

- **Actor Dataset** Includes:
  - Name
  - Last Name
  - Birthdate
- **Scraped Cast–Movie Mapping** Actor–Movie and Director–Movie relationships extracted from Wikipedia and Kinosan.mn. Used for constructing graph edges such as `acted_in` and `directed_by`.
- **Genre and Category Extraction** List of genres from scraped websites and manually curated classification rules.

3. Dataset Cleaning

A series of cleaning activities were required due to the heterogeneous and incomplete nature of Mongolian cinema sources.

- **Name normalization**: Actors and directors appear in multiple spellings on websites. Example: “ . ” vs “ - ”.
- **Duplicate removal**: Some films appear multiple times when scraped from different pages.
- **Missing values handling**: Several films lacked runtime, director, or review counts. Missing values were imputed as:
  - NULL values in the dataset
- **Standardization of release date formats**: From free-text (e.g., “2001 3- ”) to integer year values.

4. Dataset Standardisation

To support the creation of a unified Knowledge Graph schema, the datasets were standardized as follows:

- **Entity identifiers (IDs)** created for: Movies, Actors, Directors, Production Companies, and Genres.
- **Schema alignment**: All datasets aligned with a shared conceptual schema consistent with film ontologies and Schema.org patterns (`Movie`, `Person`, `Organization`, `CreativeWork`).
- **Relationship standardization**: Unified relationship labels:
  - `acted_in`
  - `directed_by`
  - `produced_by`
  - `has_genre`
  - `has_rating`
- **Attribute standardization**: Duration → integer (minutes) Release year → integer Rating → float Age rate → string

5. Discussion of Choices, Strong and Weak Points

Strengths

- **Local relevance:** The dataset reflects the actual Mongolian cinema landscape, allowing realistic KG use cases.
- **Rich metadata coverage:** The combined sources provide: title, actors, directors, genre, release year, production company,ratings, summaries, and awards — enabling meaningful graph queries.
- **Compatibility with existing global models:** The dataset can be mapped to Schema.org and Wikidata film structures to ensure interoperability.

Weaknesses

- **Heterogeneous formats:** Wikipedia and Kinosan pages differ significantly in structure, requiring extensive cleanup.
- **Incomplete data:** Many Mongolian films lack ratings, review counts, or award information.
- **Name inconsistency:** Lack of standardized transliteration or naming conventions for actors and directors.

Conclusion Despite limitations, the collected datasets (Wikipedia MN, Kinosan.mn) provide a sufficiently rich base to proceed to conceptual modeling, ontology design,and graph construction. The datasets have been cleaned, normalized, and standardized, making them ready for integration into the Mongolian Cinema Knowledge Graph.